

# From VMware to OpenStack: A Strategic Migration Blueprint

A comprehensive technical guide to planning, executing, and validating enterprise cloud migrations — from proprietary virtualization to open-source infrastructure.

CLOUD MIGRATION

OPENSTACK

INFRASTRUCTURE



# The Why: Shifting Landscapes

The virtualization market is undergoing a seismic shift. VMware's transition to mandatory subscription licensing under Broadcom ownership has sent total cost of ownership (TCO) projections soaring for enterprises of all sizes — forcing a hard look at alternatives.



## Licensing Pressure

Broadcom's acquisition of VMware introduced subscription mandates and eliminated perpetual licenses, dramatically increasing per-socket costs for many enterprise customers.



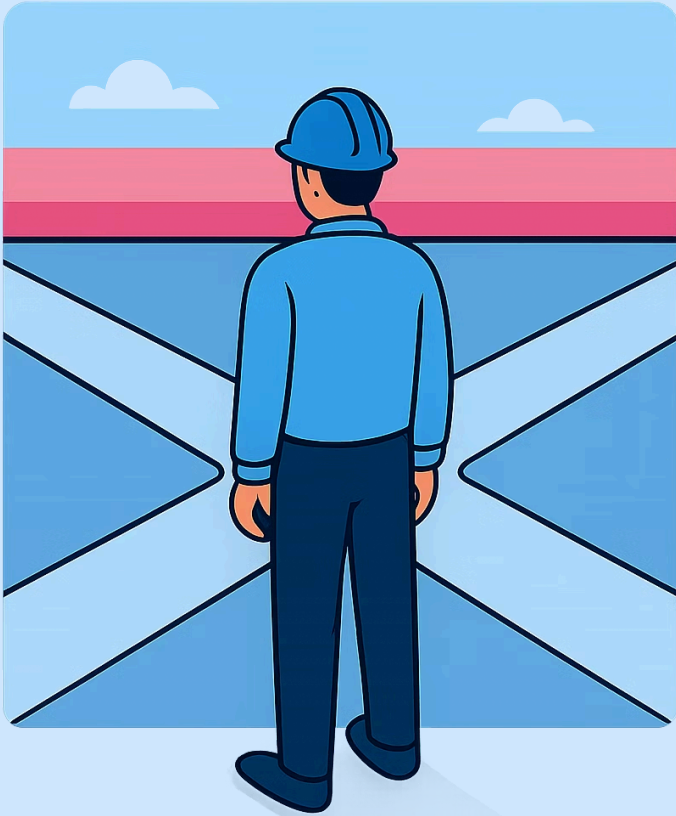
## Open-Source Momentum

OpenStack has matured into a robust, production-ready cloud platform backed by a global community. It offers API compatibility, flexibility, and zero licensing overhead.



## Legacy vs. Scale

Balancing the proven stability of existing VMware deployments against the urgent need for modern, scalable, and cost-efficient infrastructure is the central migration challenge.



# Migration Strategies: Choosing Your Path

No single migration strategy fits every workload. The industry-standard **6 R's framework** provides a structured lens for evaluating each application before any movement begins. Pair strategy selection with the right migration mode for each workload tier.

## The 6 R's Framework

01

### Rehost

Lift-and-shift with no changes

02

### Replatform

Minor optimizations for the target cloud

03

### Rearchitect

Redesign for cloud-native patterns

04

### Repurchase

Replace with SaaS or alternative

05

### Retain

Keep in place temporarily

06

### Retire

Decommission unneeded workloads

## Migration Modes



### Cold Migration

VM is fully powered off before transfer. Highest data integrity, simplest tooling — but requires scheduled downtime. Ideal for dev/test workloads.



### Warm Migration

Uses VMware's **Changed Block Tracking (CBT)** to sync incremental disk changes while the VM runs, with a brief final cutover window.



### Live Migration

Zero-downtime transfer for mission-critical workloads. Continuous replication until a near-instantaneous cutover — maximally complex but operationally essential.

# Planning: Assessment and Discovery

A successful migration begins months before the first byte is moved. Thorough discovery eliminates surprises and prevents costly rollbacks in production. Three pillars form the foundation of a solid migration plan.



## Infrastructure Audit

Systematically inventory every guest VM: vCPU count, RAM allocation, disk layout (thin vs. thick provisioned), and attached datastores. This data drives OpenStack flavor creation and capacity planning.



## Dependency Mapping

Identify inter-VM dependencies, middleware stacks, OS versions, and kernel compatibility with VirtIO drivers. Flag Windows Server and legacy Linux kernels that may require special handling.



## Digital Twin Modeling

Use tools like RVTools or Hystax discovery agents to map vCenter configurations into a structured knowledge graph — creating a digital twin that serves as the authoritative source for the target OpenStack design.

# Network and Storage Mapping

Translating VMware's networking and storage abstractions into OpenStack equivalents is one of the most nuanced aspects of migration. Precision here prevents IP conflicts, connectivity loss, and data integrity failures.

## Network Translation

### → vSwitch Port Groups → Neutron Networks

Each VMware distributed port group maps to an OpenStack Neutron network. VLAN tags, subnets, and security group rules must be replicated exactly to preserve segmentation.

### → MAC Address Preservation

Retaining the original MAC address of a migrated VM ensures ARP tables, DHCP reservations, and application-layer licensing tied to MAC remain intact post-migration.

### → Security Group Alignment

VMware firewall rules (NSX or vSphere Distributed Firewall) must be translated into equivalent Neutron security groups and router ACLs to maintain the same traffic posture.

## Storage Alignment

### Thin → Cinder Volumes

VMware thin-provisioned VMDKs map to OpenStack Cinder volumes with the same logical size, preserving storage efficiency on supported backends like Ceph or LVM-thin.

### Thick → Pre-allocated Volumes

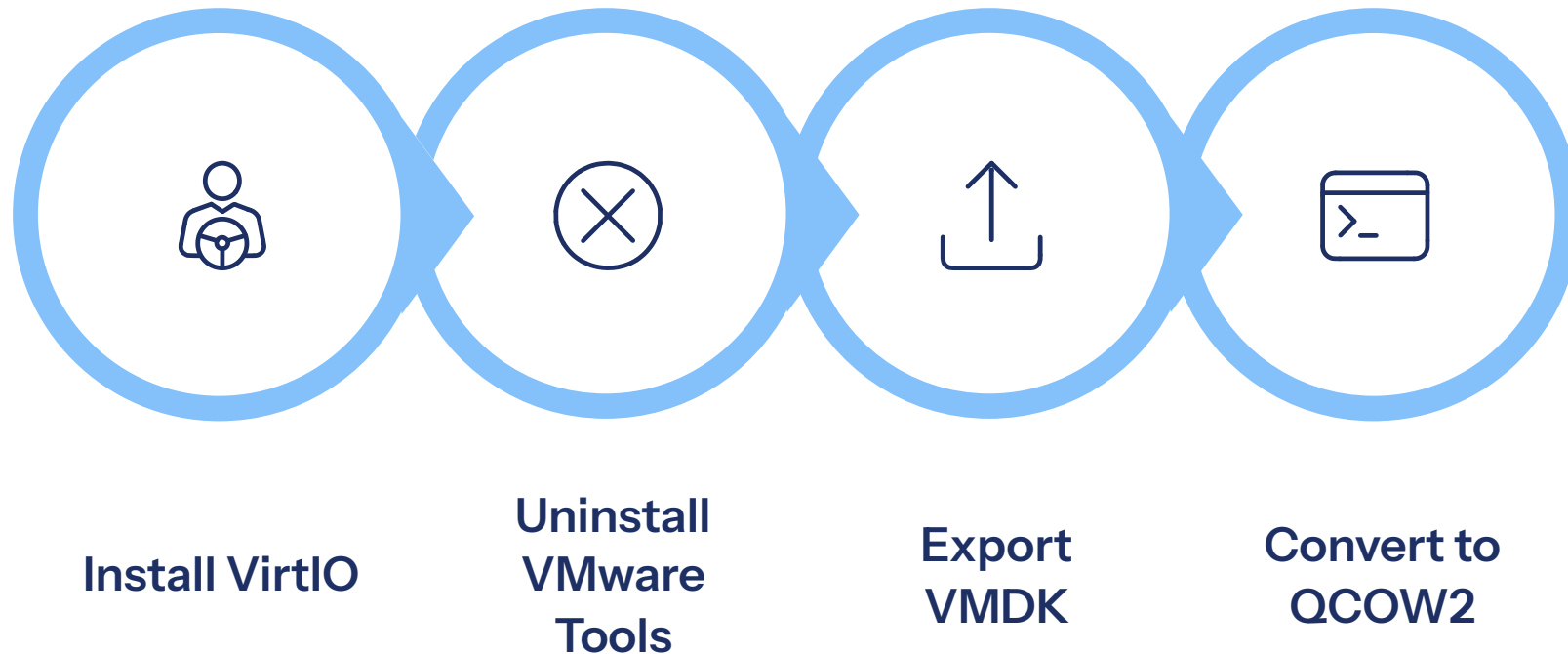
Thick-provisioned VMDKs — eager or lazy-zeroed — translate to fully pre-allocated Cinder volumes where deterministic I/O performance is required.

### Datastore → Backend Mapping

VMware datastores (VMFS, NFS, vSAN) must be mapped to Cinder storage backends with equivalent performance tiers. Volume type policies ensure QoS continuity.

# Manual Migration: The Technical Core

Manual migration provides full visibility into every transformation step — essential for understanding, troubleshooting, and validating the process at a fundamental level. Mastering this workflow is the foundation before adopting automation.



## Key Commands

```
# Convert VMDK to QCOW2
qemu-img convert \
  -f vmdk source.vmdk \
  -O qcow2 output.qcow2

# Upload to Glance
openstack image create \
  --disk-format qcow2 \
  --container-format bare \
  --file output.qcow2 \
  "migrated-vm-image"
```

## Why Each Step Matters

- **VirtIO Drivers**

OpenStack KVM hypervisor requires VirtIO for disk and network I/O. Without them, the VM fails to boot or suffers severe performance degradation.

- **Remove VMware Tools**

Leaving VMware Tools installed on a KVM guest causes system instability, failed heartbeat checks, and conflicting device drivers.

- **QCOW2 Format**

QCOW2 supports snapshots, compression, and copy-on-write — making it the native and optimal format for OpenStack Glance and Cinder.

# Automated Workflows: Hystax and Beyond

While manual migration builds foundational understanding, enterprise-scale operations demand orchestration platforms that eliminate human error, compress migration timelines, and provide audit trails for compliance.



## Hystax Acura

Hystax provides an end-to-end migration and disaster recovery platform purpose-built for VMware-to-OpenStack journeys. It deploys lightweight replication agents on each source VM, continuously streaming block-level changes to the target cloud — enabling warm and live migration at scale without manual scripting.



## Automated Discovery

Hystax auto-discovers all vCenter-managed VMs, maps their compute profiles (vCPU, RAM, disk), and builds a migration plan with **automated flavor matching** — selecting or creating the closest OpenStack flavor for each workload.



## Disk Transfer Pipelines

Block-level replication pipelines handle data transfer in the background. On cutover day, only the final delta (changed blocks since last sync) is transferred — reducing downtime windows from hours to minutes.

# Image Compatibility: Mastering the Conversion

Getting a VM image from VMware's proprietary VMDK format into an OpenStack-compatible, bootable image requires understanding the full toolchain — from format conversion to OS-level driver injection and Glance registration.

## The Conversion Toolchain

01

### qemu-img

The foundational format conversion tool. Converts between VMDK, QCOW2, RAW, VHD, and more. Supports thin-provisioned output and compression flags.

02

### virt-v2v

Red Hat's industry-standard tool for full OS conversion. Injects VirtIO drivers, removes VMware Tools, reconfigures bootloader, and validates the resulting image — all in one pass.

03

### OpenStack Glance

The image registry service for OpenStack. Accepts QCOW2, RAW, VMDK, and ISO formats. QCOW2 is preferred for its snapshot and CoW capabilities.

## Flavor Matching Logic

Once the image is uploaded, each migrated VM must be launched with an OpenStack **flavor** that meets or exceeds its original resource allocation. The matching logic follows these rules:

### vCPU Count

Flavor vCPU  $\geq$  source VM vCPU. Over-provisioning is acceptable; under-provisioning causes performance regression.

### RAM Allocation

Flavor RAM  $\geq$  source VM RAM. Factor in memory balloon driver behavior differences between ESXi and KVM.

### Root Disk Size

Flavor disk  $\geq$  QCOW2 virtual size. If no matching flavor exists, create a custom one using the OpenStack CLI before booting.



# Scaling the Effort

Migrating dozens or hundreds of VMs requires moving beyond ad-hoc scripting into structured, repeatable automation — where Ansible playbooks, CI/CD integration, and systematic validation become the backbone of the migration program.

1

## Ansible Playbooks

Red Hat OpenStack migration toolkits provide pre-built Ansible roles for bulk image conversion, flavor creation, network provisioning, and VM boot orchestration across hundreds of hosts.

2

## CI/CD Integration

Embed migration playbooks into GitLab CI or Jenkins pipelines. Each migration wave becomes a versioned, auditable pipeline run — enabling rollback, parallel execution, and drift detection.

3

## Validation & Benchmarking

Post-migration testing includes boot-time validation, network connectivity checks, application smoke tests, and performance benchmarking comparing source vs. target IOPS, throughput, and latency.

-  Treat each migration wave as a sprint: plan, execute, validate, and retrospect before the next wave begins. Incremental confidence builds operational momentum.

# The Future: A Modernized Cloud

Migration is not the destination — it is the launchpad. A successfully migrated OpenStack environment unlocks a unified control plane where virtualized VMs and containerized workloads coexist, managed through consistent APIs and open standards.

## Cloud-Native Springboard

OpenStack's Magnum and Zun services enable Kubernetes and container workloads alongside traditional VMs — accelerating the journey from "lift and shift" to truly cloud-native architecture.

## Unified Control Plane

Manage bare metal (Ironic), VMs (Nova), containers (Zun), and storage (Cinder/Swift) through a single OpenStack API surface — eliminating the sprawl of disparate management tools.

## Infrastructure Sovereignty

Reclaim control over your infrastructure stack. Open-source licensing, community-driven roadmaps, and vendor-neutral APIs eliminate single-vendor lock-in permanently.

 **Call to Action:** Begin your infrastructure audit today. Catalog every VM, map every dependency, and identify your first migration wave — the path to OpenStack sovereignty starts with a single inventory export.